Part A – Create an HTML Page Linked to manifest.json and use of API

1. Create the Project

Open Visual Studio 2026 → Create a new project → select ASP.NET Core Empty. Set the project name as StudentAPI. Select Framework: .NET 10.0 (LTS), then click Create.

2. Create the wwwroot Folder

Right-click the project → Add → New Folder. Name it wwwroot. Inside wwwroot, create:

```
wwwroot
│
├── index.html
└── manifest.json
```

3. Create manifest.json

Right-click wwwroot → Add → New Item → JSON File. Name it manifest.json and add:

```
{
  "name": "Student Management PWA",
  "short_name": "StudentPWA",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd"
}
```

The manifest.json file contains information about the PWA, such as its name, start page, display mode and theme colour.

4. Create the HTML Page

Right-click wwwroot → Add → New Item → HTML Page. Name it index.html and add:

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport"
          content="width=device-width, initial-scale=1.0">

    <title>Student Management</title>

    <!-- Link HTML page with manifest -->
    <link rel="manifest" href="/manifest.json">

</head>

<body>

    <h1>Student Management System</h1>

    <p>B.Sc. Information Technology</p>

</body>
```

```
</html>
```

The important line is:

```
<link rel="manifest" href="/manifest.json">
```

It tells the browser that this HTML page is associated with the PWA manifest file.

Part B – Create ASP.NET Core Web API

5. Create the Student Model

Right-click the project → Add → New Folder → name it Models. Right-click Models → Add → Class → name it Student.cs. Add:

```
namespace StudentAPI.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
        public string Course { get; set; }
        public string Class { get; set; }
        public string Email { get; set; }
    }
}
```

This class represents the student data.

6. Create the API Controller

Right-click the project → Add → New Folder → name it Controllers. Right-click Controllers → Add → Class → name it StudentController.cs. Add the following code using the exact student names and roll numbers provided:

```
using Microsoft.AspNetCore.Mvc;
using StudentAPI.Models;

namespace StudentAPI.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class StudentController : ControllerBase
    {
        [HttpGet]
        public IActionResult GetStudents()
        {
            var students = new List<Student>
            {
                new Student
                {
                    RollNo = 2643021,
                    Name = "SHWETA VIJAYPRAKASH GUPTA",
                    Course = "B.Sc. IT",
                    Class = "TY B.Sc. IT",
                    Email = "shweta@example.com"
                },

                new Student
                {
                    RollNo = 2643057,
                    Name = "GAYATRI SHIVKUMAR MUDALIYAR",
                    Course = "B.Sc. IT",
                    Class = "TY B.Sc. IT",
                    Email = "gayatri@example.com"
                },

                new Student
                {
                    RollNo = 2643036,
                    Name = "SUKHDEV AJAYKUMAR PARIDA",
```

```
                Course = "B.Sc. IT",
                Class = "TY B.Sc. IT",
                Email = "sukhdev@example.com"
            },

            new Student
            {
                RollNo = 2643011,
                Name = "PRANJALI JAGDISH DONGRE",
                Course = "B.Sc. IT",
                Class = "TY B.Sc. IT",
                Email = "pranjali@example.com"
            }
        };

        return Ok(students);
    }
  }
}
```

7. Configure Program.cs

Open Program.cs and replace its contents with:

```
var builder = WebApplication.CreateBuilder(args);

// Add services for Controllers
builder.Services.AddControllers();

var app = builder.Build();

// Enable static files from wwwroot
app.UseDefaultFiles();
app.UseStaticFiles();

// Enable API controllers
app.MapControllers();

app.Run();
```

app.UseDefaultFiles() allows index.html to be used as the default page. app.UseStaticFiles() allows the browser to access index.html and manifest.json from the wwwroot folder.

8. Project Structure

```
StudentAPI
│
├── Controllers
│   └── StudentController.cs
│
├── Models
│   └── Student.cs
│
├── wwwroot
│   ├── index.html
│   └── manifest.json
│
├── Program.cs
└── StudentAPI.csproj
```

9. Run the Application

Press Ctrl + F5. The application will open at https://localhost:xxxx/. You should see:

```
Student Management System

B.Sc. Information Technology
```

10. Test the Web API

The controller uses [Route("api/[controller]")] and is named StudentController. Therefore, the API URL is:

```
https://localhost:0000/api/student
```

Open this URL in the browser. The API will return the four student records in JSON format:


```
[
  {
    "rollNo": 2643021,
    "name": "SHWETA VIJAYPRAKASH GUPTA",
    "course": "B.Sc. IT",
    "class": "TY B.Sc. IT",
    "email": "shweta@example.com"
  },
  {
    "rollNo": 2643057,
    "name": "GAYATRI SHIVKUMAR MUDALIYAR",
    "course": "B.Sc. IT",
    "class": "TY B.Sc. IT",
    "email": "gayatri@example.com"
  },
  {
    "rollNo": 2643036,
    "name": "SUKHDEV AJAYKUMAR PARIDA",
    "course": "B.Sc. IT",
    "class": "TY B.Sc. IT",
    "email": "sukhdev@example.com"
  },
  {
    "rollNo": 2643011,
    "name": "PRANJALI JAGDISH DONGRE",
    "course": "B.Sc. IT",
    "class": "TY B.Sc. IT",
    "email": "pranjali@example.com"
  }
]
```